Tonglin YAN

Gif-sur-Yvette, France | tonglin.yan@universite-paris-saclay.fr | Personal Webpage | last updated in Sept. 2026

PhD student at Université Paris-Saclay working at the intersection of machine consciousness, multimodal machine learning, and virtual reality. My research focuses on integrating the computational consciousness model with large language models to study and simulate verbal and non-verbal social cognition in virtual agents. Strong background in AI, NLP, and immersive VR systems, with poster and oral presentations at international consciousness conferences.

Education

Université Paris-Saclay | Orsay, France — Oct. 2023 – present
PhD in Computational Psychology

- Advisors: **David Rudrauf** (CIAMS, Université Paris-Saclay), **Alain Finkel** (LMF, ENS Paris-Saclay), **Grégoire Sergeant-Perthuis** (CQSB, Sorbonne Université)
- Dissertation: Integration of advanced multimodal machine learning models in the Projective Consciousness Model for the study of non-verbal and verbal social behaviors in virtual reality.
- Research topics: machine consciousness, multimodal LLMs, computational modeling of consciousness, VR-based user experience.

University of Rouen Normandy | Rouen, France — Sept. 2021 – Sept. 2022
M.S. in Data Science and Engineering (Double degree with INSA Rouen Normandie)

INSA Rouen Normandie | Rouen, France — Sept. 2017 – Sept. 2022
Engineering degree in Mathematical and Software Engineering

Publications

1. **Yan, T.**, Tsukahara, J., Williford, K., Rudrauf, D., Sergeant-Perthuis, G. Formalizing Consciousness: A Comparative Mathematical Analysis of Leading Theories, Biological Cybernetics, 2026

Under Review

1. **Yan, T.**, Rudrauf, D. Looks the Same, Moves Differently: How Agent Architecture Shapes Behaviour, Experience, and Outcome in VR, 2026
2. **Yan, T.**, Sergeant-Perthuis, G., Rudrauf, D. A Hybrid PCM-LLM Architecture for Interpretable and Socially Situated Agents, 2026.
3. **Yan, T.**, Sergeant-Perthuis, G., Rudrauf, D. Belief Without Behavior: Measuring the Translation of Theory of Mind into Coordinated Social Action in Vision-Language Models, 2026.

Preprints

1. **Yan, T.**, Sergeant-Perthuis, G., Williford, K., Rudrauf, D. Integrating Machine Consciousness Simulation and LLMs Toward Verbal and Non-Verbal General Intelligence in Artificial Agents, 2025.
2. **Yan, T.**, Sergeant-Perthuis, G., Rudrauf, D. PCM-LLMS: a hybrid architecture to enhance human-like social intelligence in virtual agents, 2025.
3. Sergeant-Perthuis, G., **Yan, T.**, Ruet, N., Williford, K.,Rudrauf, D. Integrated Information Theory (IIT) with Simple Maths, 2024.

Communications

- **Yan, T.** Formalizing Consciousness: A Comparative Mathematical Analysis of Leading Theories. **7th Models of Consciousness conference (MoC7)**, Oct. 2026, Copenhagen, Denmark (talk)
- **Yan, T.** Belief Without Behavior: A Sally-Anne-Inspired Benchmark for Testing Theory of Mind Beyond Verbal Report in Vision Language Models. **MoC7**, Oct. 2026, Copenhagen, Denmark (poster)
- **Yan, T.**, Sergeant-Perthuis, G., Rudrauf, D. Minds Without Bodies: How VLMs Fail at Embodied Social Reasoning. **1st Workshop on Computational Psychology**, June 2026, ENS Paris-Saclay, France *(link to poster)*

- **Yan, T.**, Sergeant-Perthuis, G., Williford, K., Rudrauf, D. PCM-LLM: Bridging Non-Verbal Consciousness Modeling and Language Processing to Make Intelligent Social Virtual Agents Closer to Human Beings. **28th Annual Meeting of Association for the Scientific Study of Consciousness (ASSC28)**, July 2025, Crete, Greece *(link to poster)*
- Ruet, N., **Yan, T.**, Ognibene, D., Williford, K., Rudrauf, D., Sergeant-Perthuis, G. Exploring how the geometry of the representation space influences curiosity-based exploration. **ASSC27**, July 2024, Tokyo, Japan. *(link to poster)*
- Rudrauf, D., **Yan, T.**, Ruet, N., Williford, K., Sergeant-Perthuis, G. Integrated Information Theory (IIT) with Simple Maths. **ASSC27**, July 2024, Tokyo, Japan. *(link to presentation)*
- Sergeant-Perthuis, G., Ruet, N., **Yan, T.**, Williford, K., Rudrauf, D. Dualities in G-Spaces May Underly Pre-Reflective Self-Consciousness. **ASSC27**, July 2024, Tokyo, Japan. *(link to poster)*
- Zhang, L. X., **Yan, T.**, Takeda, K., Paganin, M. Optimal Transport of Hallucinatory Colors. **Post-ASSC Satellite Symposium**, July 2024, Kyoto, Japan. *(link to poster)*

Professional and Research Experiences

Algorithm Engineer, GuanYun Technology — Jan. 2023 – June 2023

- Designed and implemented a human motion capture system based on 9-axis IMU sensors.
- Developed signal processing and kinematic reconstruction algorithms for full-body motion estimation.
- Integrated motion data into Unity3D for real-time visualization and analysis in VR environments.

Research Assistant, ISIR, Sorbonne University — Mar. 2022 – Sept. 2022

- Designed an end-to-end pipeline for constructing a biology dataset.
- Fine-tuned BERT and T5 models for data-to-text generation tasks, with a 15% improvement in accuracy compared to baseline models.
- Evaluated model scalability and performance across datasets.

Research Assistant, New York University — June 2021 – Sept. 2021

- Predicted animal trajectories (ants/birds) using time-series analysis.
- Cleaned and imputed data via linear regression; extracted features with random forests.

Part-Time Assistant, Deloitte China — July 2020 – Sept. 2020

- Conducted a case study on retail markdown strategies, analyzing promotional impacts.
- Delivered client recommendations via data-driven presentations.

Academic Training

Qualia Structure Summer School | Osaka and Awaji, Japan — June 2024

- **Theme:** Explored interdisciplinary approaches to the structure of consciousness, integrating philosophy, cognitive science, neuroscience, mathematics, and artificial intelligence.

Supervision Experience

Co-supervised Le Wang (M2 intern, Mines Saint-Etienne), with David Rudrauf — 2026

Co-supervised Yannick Zelle (M1 intern, ENS), with David Rudrauf — 2025

Co-supervised Germain Poloudenny (M2 intern, Université d'Artois), with David Rudrauf — 2024

Skills

Programming: Python (PyTorch, TensorFlow), C#, Java, C++, R

Technologies: Unity 3D

Languages: Mandarin (native), English (B2), French (B2)

Interests: Tennis, Calligraphy, Bouldering, Travel